



Microsoft Sentinel Home Lab with Azure Bicep

This guide deploys a small Microsoft Sentinel lab using **Bicep**. The lab is designed for learning Infrastructure as Code (IaC), Azure Activity Logs, Kusto Query Language (KQL), Sentinel analytics rules, incident generation, and cleanup.

The lab does not deploy a virtual machine, Defender plans, a Logic App, an Azure Activity `dataConnectors` resource, or other continuously running resources. Activity Log ingestion is configured with a subscription diagnostic setting that sends records to the workspace's `AzureActivity` table.

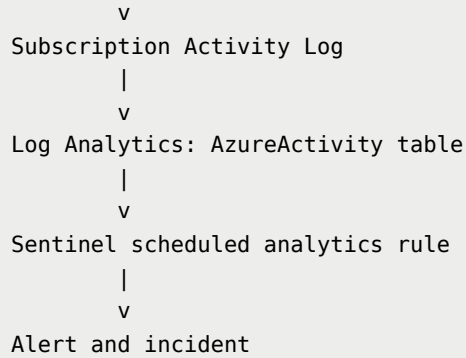
Cost warning: Microsoft Sentinel and Log Analytics can incur charges based on data ingestion and retention. Pricing, free allowances, and supported features vary by region and subscription type. Check current Azure pricing before deploying, create a budget alert, and delete the lab when finished.

1. What You Will Build

```
Azure subscription
|
|-- Resource group: rg-sentinel-homelab
|   |-- Log Analytics workspace: law-sentinel-homelab-${uniqueString(subscription().id)}
|   |-- Microsoft Sentinel onboarding state
|   |-- Scheduled analytics rules
|
|-- Subscription Activity Log diagnostic setting: send-activity-to-sentinel
|   |-- Sends eight Activity Log categories to the workspace
```

You will test the following flow:

```
Azure resource change
|
```



2. Learning Objectives

By the end of the lab, you should be able to:

- Deploy Azure resources from a Bicep template.
- Validate and redeploy Bicep safely.
- Deploy the Sentinel onboarding-state resource for a Log Analytics workspace.
- Send subscription Activity Logs to Log Analytics.
- Write and test KQL queries against `AzureActivity`.
- Create Sentinel scheduled analytics rules as code.
- Generate a controlled test event.
- Find the resulting alert and incident.
- Remove the lab without leaving subscription-level configuration behind.

3. Prerequisites

Azure requirements

- An Azure subscription where you are allowed to create resources.
- Permission to create a resource group, workspace, diagnostic setting, and Sentinel resources.
- Permission to create or remove a temporary resource group for testing.
- A browser for the Azure portal.

For the simplest setup, use an account with **Owner** or **Contributor** access plus the required Log Analytics and Microsoft Sentinel permissions. In a managed organization, ask an administrator to grant only the roles required by your organization's policy.

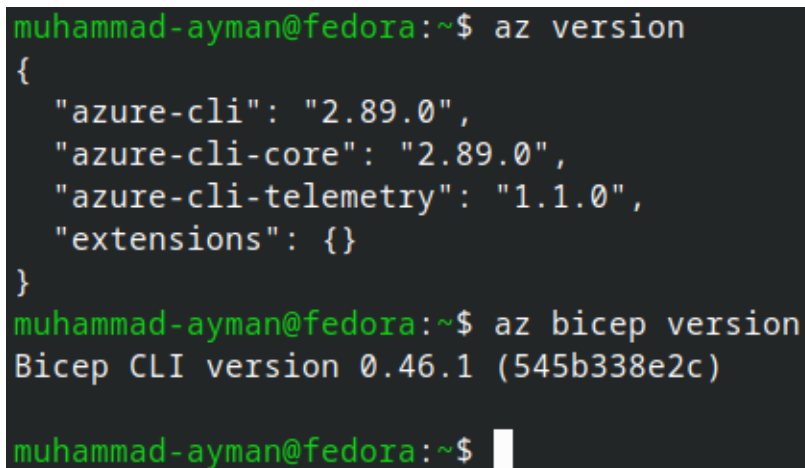
Local tools

Install:

- [Azure CLI](#)
- [Bicep CLI](#)
- A text editor, such as Visual Studio Code with the Bicep extension

Verify the tools:

```
az version
az bicep version
```



```
muhammad-ayman@fedora:~$ az version
{
  "azure-cli": "2.89.0",
  "azure-cli-core": "2.89.0",
  "azure-cli-telemetry": "1.1.0",
  "extensions": {}
}
muhammad-ayman@fedora:~$ az bicep version
Bicep CLI version 0.46.1 (545b338e2c)
muhammad-ayman@fedora:~$
```

4. Create the Lab Directory

Create a working directory and move into it:

```
mkdir sentinel-bicep-homelab
cd sentinel-bicep-homelab
```

Create a file named `main.bicep` in this directory and paste in the template from the next section.

```
muhammad-ayman@fedora:~/Downloads$ mkdir sentinel-bicep-homelab
muhammad-ayman@fedora:~/Downloads$ cd sentinel-bicep-homelab
muhammad-ayman@fedora:~/Downloads/sentinel-bicep-homelab$ touch main.bicep
muhammad-ayman@fedora:~/Downloads/sentinel-bicep-homelab$
```

5. Create the Bicep Template

The deployment uses three Bicep files and two scopes. `main.bicep` deploys at subscription scope and creates the resource group. `modules/sentinel.bicep` deploys at resource-group scope and creates the Log Analytics workspace, Sentinel onboarding state, and two scheduled analytics rules. `modules/activity-log.bicep` deploys at subscription scope and creates the subscription Activity Log diagnostic setting.

Save this as `main.bicep`:

```
targetScope = 'subscription'

@description('Azure region for the resource group and workspace.')
param location string = 'eastus'

@description('Dedicated resource group for this lab.')
param resourceGroupName string = 'rg-sentinel-homelab'

@description('Globally unique Log Analytics workspace name.')
param workspaceName string = 'law-sentinel-homelab-${uniqueString(subscription().id)}'

@description('Workspace retention in days. Increase only when needed.')
@minValue(30)
param retentionInDays int = 30
```

```

resource labResourceGroup 'Microsoft.Resources/resourceGroups
@2023-07-01' = {
  name: resourceGroupName
  location: location
  tags: {
    environment: 'homelab'
    managedBy: 'bicep'
    purpose: 'microsoft-sentinel-learning'
  }
}

module workspace 'modules/workspace.bicep' = {
  name: 'sentinel-workspace'
  scope: labResourceGroup
  params: {
    location: location
    workspaceName: workspaceName
    retentionInDays: retentionInDays
  }
}

output resourceGroupId string = labResourceGroup.id
output resourceGroupName string = labResourceGroup.name
output workspaceResourceId string = workspace.outputs.workspa
ceResourceId
output workspaceName string = workspace.outputs.workspaceName

```

```

muhammad-ayman@fedora:~/Downloads/sentinel-bicep-homelab$ nano main.bicep

```

```
GNU nano 8.7.1 main.bicep
targetScope = 'subscription'

@description('Azure region for the resource group and workspace.')
param location string = 'eastus'

@description('Dedicated resource group for this lab.')
param resourceGroupName string = 'rg-sentinel-homelab'

@description('Globally unique Log Analytics workspace name.')
param workspaceName string = 'law-sentinel-homelab-${uniqueString(subscription().id)}'

@description('Workspace retention in days. Increase only when needed.')
@minValue(30)
param retentionInDays int = 30

@description('Name of the subscription Activity Log diagnostic setting.')
param diagnosticSettingName string = 'send-activity-to-sentinel'

resource labResourceGroup 'Microsoft.Resources/resourceGroups@2023-07-01' = {
  name: resourceGroupName
  location: location
  tags: {
    environment: 'homelab'
    managedBy: 'bicep'
    purpose: 'microsoft-sentinel-learning'
  }
}

resource workspace 'Microsoft.OperationalInsights/workspaces@2023-09-01' = {
  scope: labResourceGroup
  name: workspaceName
  location: location
  tags: {
    environment: 'homelab'
    managedBy: 'bicep'
    purpose: 'microsoft-sentinel-learning'
  }
  properties: {
    sku: {
      name: 'PerGB2018'
    }
  }
}
```

Create the module directory and save this as `sentinel.bicep`:

```
targetScope = 'subscription'

@description('Azure resource group containing the Log Analytics workspace.')
param resourceGroupName string = 'rg-sentinel-homelab'

@description('Existing Log Analytics workspace name.')
param workspaceName string = 'law-sentinel-homelab-${uniqueString(subscription().id)}'

@description('Name of the subscription Activity Log diagnostic setting.')
param diagnosticSettingName string = 'send-activity-to-sentinel'

resource labResourceGroup 'Microsoft.Resources/resourceGroups@2023-07-01' existing = {
  name: resourceGroupName
}
```

```

module sentinelLab 'modules/sentinel.bicep' = {
  name: 'sentinel-rules'
  scope: labResourceGroup
  params: {
    workspaceName: workspaceName
  }
}

module activityLog 'modules/activity-log.bicep' = {
  name: 'sentinel-activity-log'
  scope: subscription()
  params: {
    diagnosticSettingName: diagnosticSettingName
    workspaceResourceId: sentinelLab.outputs.workspaceResourceId
  }
}

output workspaceResourceId string = sentinelLab.outputs.workspaceResourceId
output workspaceName string = sentinelLab.outputs.workspaceName
output diagnosticSettingResourceId string = activityLog.outputs.diagnosticSettingResourceId

```

Create `modules/activity-log.bicep` with the following content:

```

targetScope = 'subscription'

@description('Name of the subscription Activity Log diagnostic setting.')
param diagnosticSettingName string

@description('Resource ID of the Log Analytics workspace receiving Activity Logs.')

```

```

param workspaceResourceId string

resource activityDiagnosticSetting 'Microsoft.Insights/diagnosticSettings@2021-05-01-preview' = {
  scope: subscription()
  name: diagnosticSettingName
  // The subscription diagnostic-settings API persists this as Global.
  // The current Bicep type does not expose the valid location property.
  location: 'global'
  properties: {
    workspaceId: workspaceResourceId
    logs: [
      {
        category: 'Administrative'
        enabled: true
      }
      {
        category: 'Security'
        enabled: true
      }
      {
        category: 'Policy'
        enabled: true
      }
      {
        category: 'ServiceHealth'
        enabled: true
      }
      {
        category: 'ResourceHealth'
        enabled: true
      }
      {
        category: 'Alert'

```



```

        enabled: true
      }
    {
      category: 'Recommendation'
      enabled: true
    }
    {
      category: 'Autoscale'
      enabled: true
    }
  ]
}

```

```

output diagnosticSettingResourceId string = activityDiagnosticSetting.id

```

Create `modules/sentinel.bicep` with the following content:

```

targetScope = 'resourceGroup'

@description('Globally unique Log Analytics workspace name.')
param workspaceName string

resource workspace 'Microsoft.OperationalInsights/workspaces@2023-09-01' existing = {
  name: workspaceName
}

resource sentinel 'Microsoft.SecurityInsights/onboardingStates@2025-09-01' = {
  name: 'default'
  scope: workspace
  properties: {
    customerManagedKey: false
  }
}

```

```

}

resource networkChangeRule 'Microsoft.SecurityInsights/alertRules@2023-02-01' = {
  name: guid(subscription().id, 'sentinel-network-change-rule')
  scope: workspace
  dependsOn: [
    sentinel
  ]
  kind: 'Scheduled'
  properties: {
    displayName: 'Lab - Network security group changed'
    description: 'Detects creation, update, or deletion of a network security group.'
    severity: 'Low'
    enabled: true
    query: '''
      AzureActivity
      | where CategoryValue == "Administrative"
      | where ResourceProviderValue =~ "Microsoft.Network"
      | where OperationNameValue has_any ("networkSecurityGroups", "securityGroups")
      | where ActivityStatusValue == "Success"
      | project TimeGenerated, Caller, OperationNameValue, ResourceGroup, ResourceId, ActivityStatusValue
    '''
    queryFrequency: 'PT5M'
    queryPeriod: 'PT10M'
    triggerOperator: 'GreaterThan'
    triggerThreshold: 0
    suppressionEnabled: false
    suppressionDuration: 'PT5H'
    incidentConfiguration: {
      createIncident: true
      groupingConfiguration: {

```

```

        enabled: false
        lookbackDuration: 'PT5H'
        matchingMethod: 'AllEntities'
        reopenClosedIncident: false
    }
}
tactics: [
    'DefenseEvasion'
]
}
}

resource policyChangeRule 'Microsoft.SecurityInsights/alertRules@2023-02-01' = {
    name: guid(subscription().id, 'sentinel-policy-change-rule')
    scope: workspace
    dependsOn: [
        sentinel
    ]
    kind: 'Scheduled'
    properties: {
        displayName: 'Lab - Azure Policy assignment changed'
        description: 'Detects a successful Azure Policy assignment write or delete operation.'
        severity: 'Medium'
        enabled: true
        query: '''
            AzureActivity
            | where CategoryValue == "Administrative"
            | where OperationNameValue has "policyAssignments"
            | where ActivityStatusValue == "Success"
            | project TimeGenerated, Caller, OperationNameValue, ResourceGroup, ResourceId, ActivityStatusValue
            ...
        '''
        queryFrequency: 'PT5M'
    }
}

```

```

    queryPeriod: 'PT10M'
    triggerOperator: 'GreaterThan'
    triggerThreshold: 0
    suppressionEnabled: false
    suppressionDuration: 'PT5H'
    incidentConfiguration: {
      createIncident: true
      groupingConfiguration: {
        enabled: false
        lookbackDuration: 'PT5H'
        matchingMethod: 'AllEntities'
        reopenClosedIncident: false
      }
    }
    tactics: [
      'DefenseEvasion'
    ]
  }
}

```

```

output workspaceResourceId string = workspace.id
output workspaceName string = workspace.name

```

Create `modules/workspace.bicep` with the following content:

```

targetScope = 'resourceGroup'

@description('Azure region for the workspace.')
param location string

@description('Globally unique Log Analytics workspace name.')
param workspaceName string

@description('Workspace retention in days.')
@minValue(30)
param retentionInDays int = 30

```

```

var labTags = {
  environment: 'homelab'
  managedBy: 'bicep'
  purpose: 'microsoft-sentinel-learning'
}

resource workspace 'Microsoft.OperationalInsights/workspaces@2023-09-01' = {
  name: workspaceName
  location: location
  tags: labTags
  properties: {
    sku: {
      name: 'PerGB2018'
    }
    retentionInDays: retentionInDays
    features: {
      enableLogAccessUsingOnlyResourcePermissions: true
    }
  }
}

output workspaceResourceId string = workspace.id
output workspaceName string = workspace.name

```

Notes about the template

- `targetScope = 'subscription'` is required in `main.bicep` and `modules/activity-log.bicep` because the resource group and diagnostic setting are subscription-scoped.
- The module is required because Bicep does not allow a subscription-scope file to directly deploy a resource-group-scoped workspace.
- Sentinel onboarding and analytics rules use `scope: workspace`; Microsoft Sentinel is associated with a Log Analytics workspace, not directly with the

resource group.

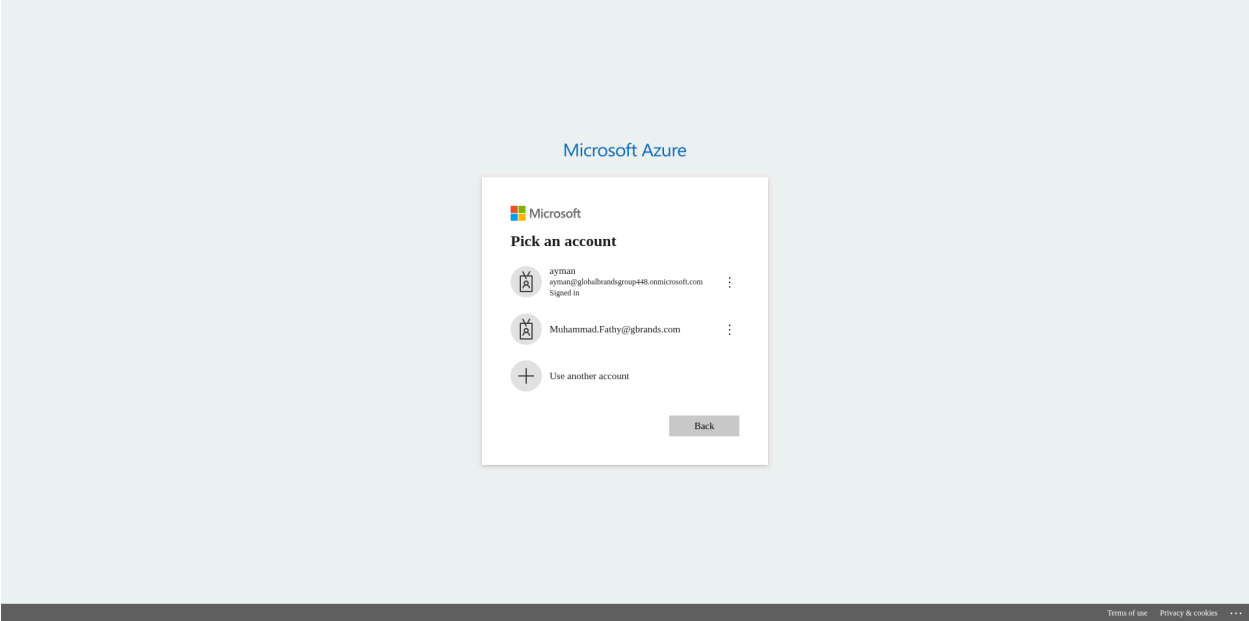
- `uniqueString(subscription().id)` prevents a workspace-name collision when the default is used.
- The workspace retention is set to 30 days. This is a minimum-style lab setting, not a production recommendation.
- Activity Log ingestion and Sentinel behavior can take several minutes. Do not expect an incident immediately after an operation.
- The exact Activity Log operation names can differ by resource provider. The KQL queries intentionally use `has` and `has_any`; inspect actual events and refine them for your subscription.
- The deployment creates a new workspace by default. It does not reuse an existing workspace such as `sentinellabaml1304768739` unless you pass that name as the `workspaceName` parameter.
- The Bicep deployment does not install the Azure Activity solution from Content hub or create a separate `Microsoft.SecurityInsights/dataConnectors` resource. The diagnostic setting is the configuration that forwards Activity Logs to Log Analytics.
- The `policyChangeRule` is a Sentinel detection rule. It is not an Azure Policy assignment or an automatic remediation policy.

6. Sign In and Select the Subscription

Sign in to Azure:

```
az login
```

```
muhammad-ayman@fedora:~/Downloads/sentinel-bicep-homelab$ az login
A web browser has been opened at https://login.microsoftonline.com/organizations/oauth2/v2.0/authorize. Please continue the login in the web browser. If no web browser is available or if the web browser fails to open, use device code flow with 'az login --use-device-code'.
```



You have logged into Microsoft Azure!
You can close this window, or we will redirect you to the [Azure CLI documentation](#) in 1 minute.

```
muhammad-ayman@fedora:~/Downloads/sentinel-bicep-homelab$ az login
A web browser has been opened at https://login.microsoftonline.com/organizations/oauth2/v2.0/authorize. Please continue the login in the web browser. If no web browser is available or if the web browser fails to open, use device code flow with 'az login --use-device-code'.

Retrieving tenants and subscriptions for the selection...

[Tenant and subscription selection]

No      Subscription name      Subscription ID      Tenant
-----
[1] *   Ayman-Visual Studio Enterprise Su...  1132ce03-611b-4164-9042-413aeb38825b  global brands group
[2]     Lasheen- MPN                1132ce03-611b-4164-9042-413aeb38825b  global brands group
[3]     Visual Studio Enterprise Subscrip...  1132ce03-611b-4164-9042-413aeb38825b  global brands group
[4]     Visual Studio Enterprise Subscrip...  1132ce03-611b-4164-9042-413aeb38825b  global brands group
[5]     Zeyad subscription - MPN            1132ce03-611b-4164-9042-413aeb38825b  global brands group

The default is marked with an *; the default tenant is 'global brands group' and subscription is 'Lasheen- MPN' (1132ce03-611b-4164-9042-413aeb38825b).

Select a subscription and tenant (Type a number or Enter for no changes): 1
```

```
muhammad-ayman@fedora:~/Downloads/sentinel-bicep-homelab$ az login
A web browser has been opened at https://login.microsoftonline.com/organizations/oauth2/v2.0/authorize. Please continue the login in the web browser. If no web browser is available or if the web browser fails to open, use device code flow with 'az login --use-device-code'.

Retrieving tenants and subscriptions for the selection...

[Tenant and subscription selection]

No      Subscription name      Subscription ID      Tenant
-----
[1] *   Ayman-Visual Studio Enterprise Su...  1132ce03-611b-4164-9042-413aeb38825b  global brands group
[2]     Lasheen- MPN                1132ce03-611b-4164-9042-413aeb38825b  global brands group
[3]     Visual Studio Enterprise Subscrip...  1132ce03-611b-4164-9042-413aeb38825b  global brands group
[4]     Visual Studio Enterprise Subscrip...  1132ce03-611b-4164-9042-413aeb38825b  global brands group
[5]     Zeyad subscription - MPN            1132ce03-611b-4164-9042-413aeb38825b  global brands group

The default is marked with an *; the default tenant is 'global brands group' and subscription is 'Lasheen- MPN' (1132ce03-611b-4164-9042-413aeb38825b).

Select a subscription and tenant (Type a number or Enter for no changes): 1

Tenant: global brands group
Subscription: Ayman-Visual Studio Enterprise Subscription - MPN (1132ce03-611b-4164-9042-413aeb38825b)

[Announcements]
With the new Azure CLI login experience, you can select the subscription you want to use more easily. Learn more about it and its configuration at https://go.microsoft.com/fwlink/?linkid=2271236

If you encounter any problem, please open an issue at https://aka.ms/azclibug

[Warning] The login output has been updated. Please be aware that it no longer displays the full list of available subscriptions by default.

muhammad-ayman@fedora:~/Downloads/sentinel-bicep-homelab$
```

Confirm the active subscription:

```
az account show --query '{name:name,id:id,tenantId:tenantId}'
--output table
```



```
muhammad-ayman@fedora:~/Downloads/sentinel-bicep-homelab$ az account show --query '{name:name,id:id,tenantId:tenantId}' --output table
Name                               TenantId
-----
Ayman-Visual Studio Enterprise Subscription - MPN  7b9e42e1-91d1-48a1-bf00-000000000000
muhammad-ayman@fedora:~/Downloads/sentinel-bicep-homelab$
```

Never run this lab against a production subscription unless the subscription owner has approved the deployment.

7. Validate the Bicep Template

Run the Bicep linter/build step:

```
az bicep build --file main.bicep
az bicep build --file sentinel.bicep
```

Validate the subscription deployment without creating resources:

```
az deployment sub validate \
  --name sentinel-homelab-validation \
  --location eastus \
  --template-file main.bicep \
  --parameters location=eastus
```

If validation fails, read the first reported resource and property error. Common causes are insufficient permissions, a region that does not support the selected service, or an API version mismatch in an older CLI installation.

```

muhammad-ayan@fedora:~/Downloads/sentinel-bicep-homelab$ az bicep build --file main.bicep
muhammad-ayan@fedora:~/Downloads/sentinel-bicep-homelab$ az deployment sub validate \
  --name sentinel-homelab-validation \
  --location eastus \
  --template-file main.bicep \
  --parameters location=eastus
{
  "error": null,
  "id": "/subscriptions/60ba34d6-e952-486b-b30d-a3109360cd74/providers/Microsoft.Resources/deployments/sentinel-homelab-validation",
  "location": "eastus",
  "name": "sentinel-homelab-validation",
  "properties": {
    "correlationId": "96daa23f-8273-4919-92d1-4e16e98dc52",
    "debugSetting": null,
    "dependencies": [
      {
        "dependsOn": [
          {
            "id": "/subscriptions/60ba34d6-e952-486b-b30d-a3109360cd74/resourceGroups/rg-sentinel-homelab/providers/Microsoft.Resources/deployments/sentinel-lab-resources",
            "resourceGroup": "rg-sentinel-homelab",
            "resourceName": "sentinel-lab-resources",
            "resourceType": "Microsoft.Resources/deployments"
          },
          {
            "apiVersion": "2025-04-01",
            "id": "/subscriptions/60ba34d6-e952-486b-b30d-a3109360cd74/resourceGroups/rg-sentinel-homelab/providers/Microsoft.Resources/deployments/sentinel-lab-resources",
            "resourceGroup": "rg-sentinel-homelab",
            "resourceName": "sentinel-lab-resources",
            "resourceType": "Microsoft.Resources/deployments"
          }
        ],
        "id": "/subscriptions/60ba34d6-e952-486b-b30d-a3109360cd74/providers/Microsoft.Insights/diagnosticSettings/send-activity-to-sentinel",
        "resourceName": "send-activity-to-sentinel",
        "resourceType": "Microsoft.Insights/diagnosticSettings"
      }
    ],
    "dependsOn": [
      {
        "id": "/subscriptions/60ba34d6-e952-486b-b30d-a3109360cd74/resourceGroups/rg-sentinel-homelab",
        "resourceName": "rg-sentinel-homelab",
        "resourceType": "Microsoft.Resources/resourceGroups"
      }
    ]
  },
  "id": "/subscriptions/60ba34d6-e952-486b-b30d-a3109360cd74/resourceGroups/rg-sentinel-homelab/providers/Microsoft.Resources/deployments/sentinel-lab-resources",

```

8. Deploy the Lab

Deployment Stage 1: Resource Group and Workspace

The first stage creates:

- The `rg-sentinel-homelab` resource group
- The `law-sentinel-homelab-g32ur33aju6wk` Log Analytics workspace

Run:

```

az deployment sub create \
  --name sentinel-homelab \
  --location eastus \
  --template-file main.bicep \
  --parameters location=eastus

```

```

muhammad-ayman@fedora:~/Downloads/sentinel-bicep-homelab$ az deployment sub create \
> --name sentinel-homelab \
> --location eastus \
> --template-file main.bicep \
> --parameters location=eastus
{
  "id": "/subscriptions/60ba34d6-e952-486b-b30d-a3109360cd74/providers/Microsoft.Resources/deployments/sentinel-homelab",
  "location": "eastus",
  "name": "sentinel-homelab",
  "properties": {
    "correlationId": "dc068712-a9e5-427f-a450-827afbe42bd1",
    "debugSetting": null,
    "dependencies": [
      {
        "dependsOn": [
          {
            "id": "/subscriptions/60ba34d6-e952-486b-b30d-a3109360cd74/resourceGroups/rg-sentinel-homelab/providers/Microsoft.Resources/deployments/sentinel-lab-resources",
            "resourceGroup": "rg-sentinel-homelab",
            "resourceName": "sentinel-lab-resources",
            "resourceType": "Microsoft.Resources/deployments"
          },
          {
            "apiVersion": "2025-04-01",
            "id": "/subscriptions/60ba34d6-e952-486b-b30d-a3109360cd74/resourceGroups/rg-sentinel-homelab/providers/Microsoft.Resources/deployments/sentinel-lab-resources",
            "resourceGroup": "rg-sentinel-homelab",
            "resourceName": "sentinel-lab-resources",
            "resourceType": "Microsoft.Resources/deployments"
          }
        ],
        "id": "/subscriptions/60ba34d6-e952-486b-b30d-a3109360cd74/providers/Microsoft.Insights/diagnosticSettings/send-activity-to-sentinel",
        "resourceName": "send-activity-to-sentinel",
        "resourceType": "Microsoft.Insights/diagnosticSettings"
      },
      {
        "dependsOn": [
          {
            "id": "/subscriptions/60ba34d6-e952-486b-b30d-a3109360cd74/resourceGroups/rg-sentinel-homelab",
            "resourceName": "rg-sentinel-homelab",
            "resourceType": "Microsoft.Resources/resourceGroups"
          }
        ],
        "id": "/subscriptions/60ba34d6-e952-486b-b30d-a3109360cd74/resourceGroups/rg-sentinel-homelab/providers/Microsoft.Resources/deployments/sentinel-lab-resources",
        "resourceGroup": "rg-sentinel-homelab",
        "resourceName": "sentinel-lab-resources",
      }
    ]
  }
}

```

Wait for Azure Propagation

Wait approximately 30–60 seconds before starting the second stage.

The workspace may be provisioned successfully by Azure Resource Manager while Microsoft Sentinel is still registering the workspace internally. This delay prevents the analytics rule validation error.

Deployment Stage 2: Microsoft Sentinel Resources

The second stage creates:

- Microsoft Sentinel onboarding
- The network security group change analytics rule
- The Azure Policy assignment change analytics rule
- The subscription Activity Log diagnostic setting

Run:

```

az deployment sub create \
  --name sentinel-resources \
  --location eastus \
  --template-file sentinel.bicep \

```

```
--parameters \
  resourceGroupName=rg-sentinel-homelab \
  workspaceName=law-sentinel-homelab-g32ur33aju6wk
```

Wait for the deployment to finish. Record the output workspace name:

```
az deployment sub show \
  --name sentinel-homelab \
  --query properties.outputs \
  --output json
```

```
muhammad-ayman@fedora:~/Downloads/sentinel-bicep-homelab$ az deployment sub show \
> --name sentinel-homelab \
> --query properties.outputs \
> --output json
{
  "resourceGroupId": {
    "type": "String",
    "value": "/subscriptions/60ba34d6-e952-486b-b30d-a3109360cd74/resourceGroups/rg-sentinel-homelab"
  },
  "workspaceName": {
    "type": "String",
    "value": "law-sentinel-homelab-g32ur33aju6wk"
  },
  "workspaceResourceId": {
    "type": "String",
    "value": "/subscriptions/60ba34d6-e952-486b-b30d-a3109360cd74/resourceGroups/rg-sentinel-homelab/providers/Microsoft.OperationalInsights/workspaces/law-sentinel-homelab-g32ur33aju6wk"
  }
}
```

Verify the resource group and resources:

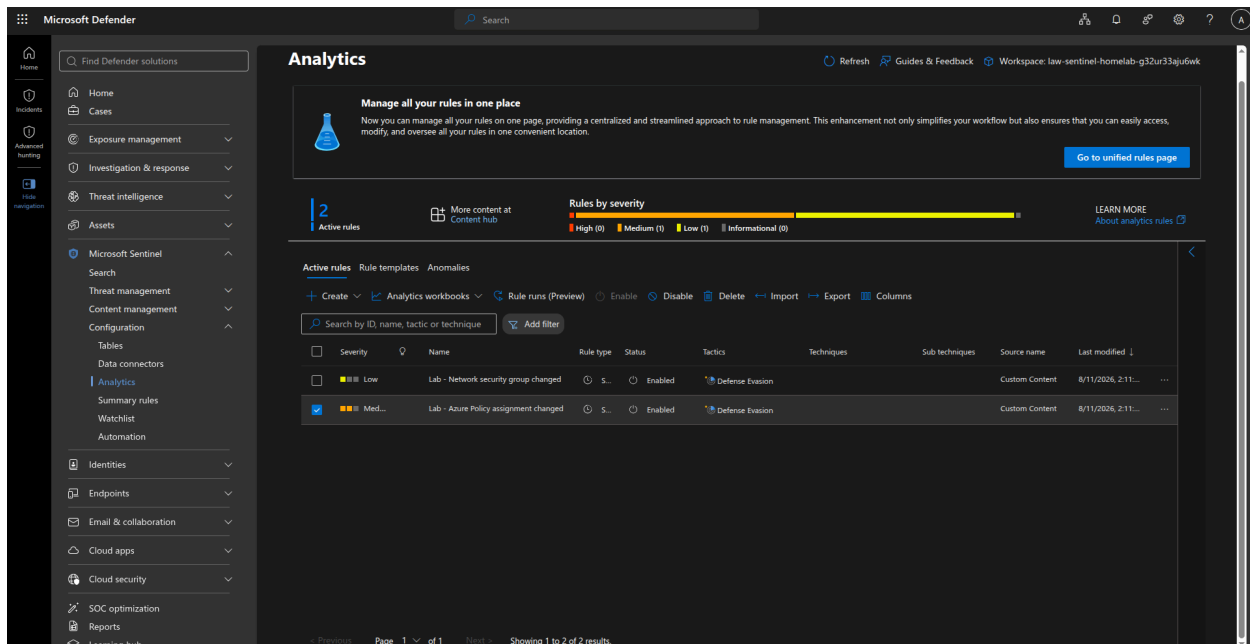
```
az group show --name rg-sentinel-homelab --output table
az resource list --resource-group rg-sentinel-homelab --output
t table
```

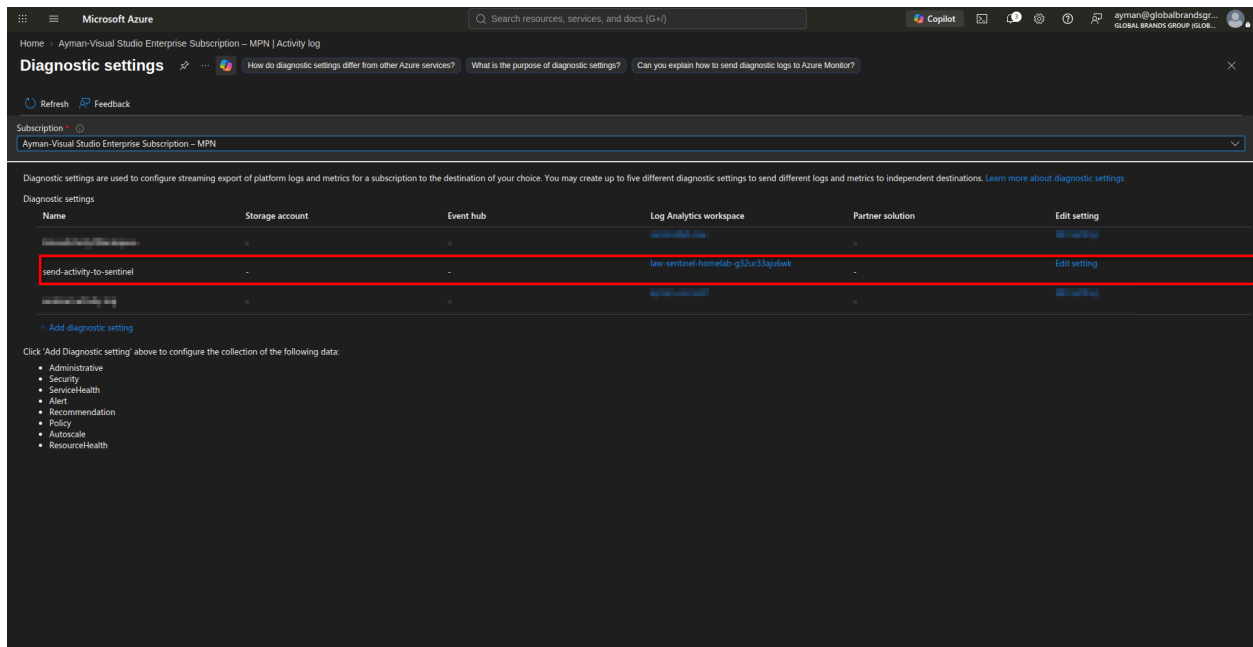
```
muhammad-ayman@fedora:~/Downloads/sentinel-bicep-homelab$ az group show --name rg-sentinel-homelab --output table
Location      Name
-----
eastus        rg-sentinel-homelab
muhammad-ayman@fedora:~/Downloads/sentinel-bicep-homelab$ az resource list --resource-group rg-sentinel-homelab --output table
Name                                     ResourceGroup      Location      Type                                     Status
-----
law-sentinel-homelab-g32ur33aju6wk      rg-sentinel-homelab eastus        Microsoft.OperationalInsights/workspaces Succeeded
SecurityInsights(law-sentinel-homelab-g32ur33aju6wk) rg-sentinel-homelab eastus        Microsoft.OperationsManagement/solutions Succeeded
```

9. Check Sentinel in the Azure Portal

1. Open the Azure portal.
2. Search for **Microsoft Sentinel**.
3. Open the workspace named by the deployment output.

4. Confirm that the workspace is available in Sentinel. It might be shown as not onboarded but it is.
5. Open **Analytics** and confirm the two rules beginning with **Lab -** are enabled.
6. Open the subscription's **Activity log > Diagnostic settings** and confirm **send-activity-to-sentinel** targets the deployed workspace.
7. If the workspace does not appear in the old Azure portal Sentinel list, check the Microsoft Defender portal at <https://security.microsoft.com> under **Settings > Microsoft Sentinel > Workspaces**. Newly created workspaces can be redirected there.

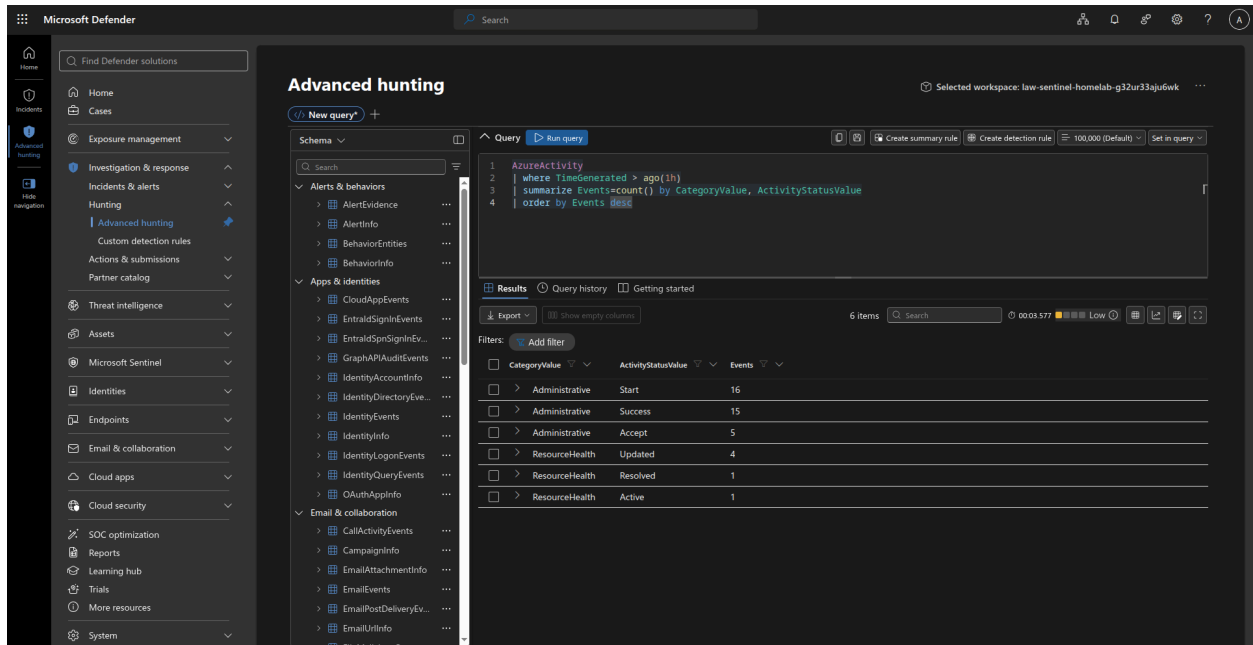




10. Verify Activity Log Ingestion with KQL

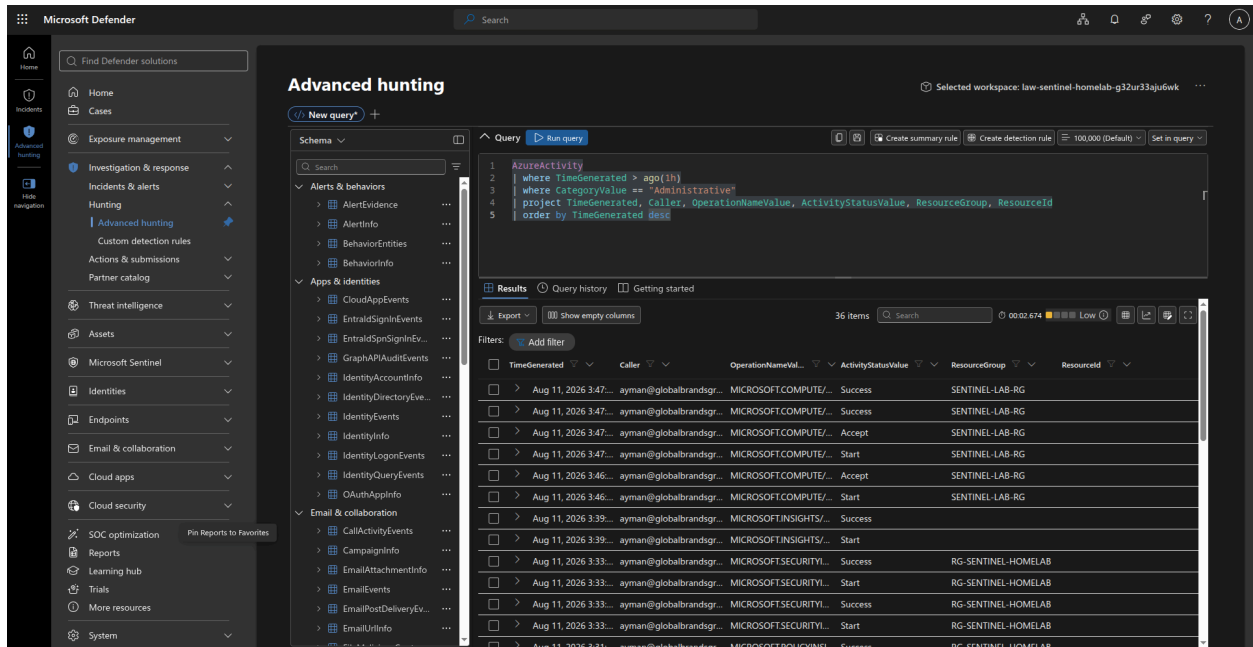
In the Sentinel workspace, open **Logs** and run:

```
AzureActivity
| where TimeGenerated > ago(1h)
| summarize Events=count() by CategoryValue, ActivityStatusValue
| order by Events desc
```



Show recent administrative events:

```
AzureActivity
| where TimeGenerated > ago(1h)
| where CategoryValue == "Administrative"
| project TimeGenerated, Caller, OperationNameValue, ActivityStatusValue, ResourceGroup,
ResourceId
| order by TimeGenerated desc
```



If there are no rows, confirm that the subscription diagnostic setting exists and wait up to 15 minutes before testing again.

11. Generate a Safe Test Event

Use a separate temporary resource group. This avoids changing an existing application resource.

Create the test resource group:

```
az group create \
  --name rg-sentinel-test-events \
  --location eastus \
  --tags environment=homelab purpose=sentinel-test
```



```

muhammad-ayman@fedora:~/Downloads/sentinel-bicep-homelab$ az group create \
> --name rg-sentinel-test-events \
> --location eastus \
> --tags environment=homelab purpose=sentinel-test
{
  "id": "/subscriptions/60ba34d6-e952-486b-b30d-a3109360cd74/resourceGroups/rg-sentinel-test-events",
  "location": "eastus",
  "managedBy": null,
  "name": "rg-sentinel-test-events",
  "properties": {
    "provisioningState": "Succeeded"
  },
  "tags": {
    "environment": "homelab",
    "purpose": "sentinel-test"
  },
  "type": "Microsoft.Resources/resourceGroups"
}
muhammad-ayman@fedora:~/Downloads/sentinel-bicep-homelab$

```

Create a temporary network security group:

```

az network nsg create \
  --resource-group rg-sentinel-test-events \
  --name nsg-sentinel-test

```

```

muhammad-ayman@fedora:~/Downloads/sentinel-bicep-homelab$ az network nsg create \
> --resource-group rg-sentinel-test-events \
> --name nsg-sentinel-test
{
  "NewNSG": {
    "defaultSecurityRules": [
      {
        "access": "Allow",
        "description": "Allow inbound traffic from all VMs in VNET",
        "destinationAddressPrefix": "VirtualNetwork",
        "destinationAddressPrefixes": [],
        "destinationPortRange": "**",
        "destinationPortRanges": [],
        "direction": "Inbound",
        "etag": "W/\"bb7d309f-a01a-434f-8b63-abb6837e11e\"",
        "id": "/subscriptions/60ba34d6-e952-486b-b30d-a3109360cd74/resourceGroups/rg-sentinel-test-events/providers/Microsoft.Network/networkSecurityGroups/nsg-sentinel-test/defaultSecurityRules/AllowVnetInBound",
        "name": "AllowVnetInBound",
        "priority": 65000,
        "protocol": "**",
        "provisioningState": "Succeeded",
        "resourceGroup": "rg-sentinel-test-events",
        "sourceAddressPrefix": "VirtualNetwork",
        "sourceAddressPrefixes": [],
        "sourcePortRange": "**",
        "sourcePortRanges": [],
        "type": "Microsoft.Network/networkSecurityGroups/defaultSecurityRules"
      },
      {
        "access": "Allow",
        "description": "Allow inbound traffic from azure load balancer",
        "destinationAddressPrefix": "**",
        "destinationAddressPrefixes": [],
        "destinationPortRange": "**",
        "destinationPortRanges": [],
        "direction": "Inbound",
        "etag": "W/\"bb7d309f-a01a-434f-8b63-abb6837e11e\"",
        "id": "/subscriptions/60ba34d6-e952-486b-b30d-a3109360cd74/resourceGroups/rg-sentinel-test-events/providers/Microsoft.Network/networkSecurityGroups/nsg-sentinel-test/defaultSecurityRules/AllowAzureLoadBalancerInBound",
        "name": "AllowAzureLoadBalancerInBound",
        "priority": 65001,
        "protocol": "**",
        "provisioningState": "Succeeded",
        "resourceGroup": "rg-sentinel-test-events",

```

Update it with a harmless description:

```
az network nsg update \
  --resource-group rg-sentinel-test-events \
  --name nsg-sentinel-test \
  --tags testChange=updated
```

```
muhammad-ayman@fedora:~/Downloads/sentinel-bicep-homelab$ az network nsg update \
> --resource-group rg-sentinel-test-events \
> --name nsg-sentinel-test \
> --tags testChange=updated
{
  "defaultSecurityRules": [
    {
      "access": "Allow",
      "description": "Allow inbound traffic from all VMs in VNET",
      "destinationAddressPrefix": "VirtualNetwork",
      "destinationAddressPrefixes": [],
      "destinationPortRange": "**",
      "destinationPortRanges": [],
      "direction": "Inbound",
      "etag": "W/\"7b17692a-4cc2-4e03-a7a6-238bd7456ce1\"",
      "id": "/subscriptions/60ba34d6-e952-486b-b30d-a3109360cd74/resourceGroups/rg-sentinel-test-events/providers/Microsoft.Network/networkSecurityGroups/nsg-sentinel-test/defaultSecurityRules/AllowVnetInBound",
      "name": "AllowVnetInBound",
      "priority": 65000,
      "protocol": "**",
      "provisioningState": "Succeeded",
      "resourceGroup": "rg-sentinel-test-events",
      "sourceAddressPrefix": "VirtualNetwork",
      "sourceAddressPrefixes": [],
      "sourcePortRange": "**",
      "sourcePortRanges": [],
      "type": "Microsoft.Network/networkSecurityGroups/defaultSecurityRules"
    },
    {
      "access": "Allow",
      "description": "Allow inbound traffic from azure load balancer",
      "destinationAddressPrefix": "**",
      "destinationAddressPrefixes": [],
      "destinationPortRange": "**",
      "destinationPortRanges": [],
      "direction": "Inbound",
      "etag": "W/\"7b17692a-4cc2-4e03-a7a6-238bd7456ce1\"",
      "id": "/subscriptions/60ba34d6-e952-486b-b30d-a3109360cd74/resourceGroups/rg-sentinel-test-events/providers/Microsoft.Network/networkSecurityGroups/nsg-sentinel-test/defaultSecurityRules/AllowAzureLoadBalancerInBound",
      "name": "AllowAzureLoadBalancerInBound",
      "priority": 65001,
      "protocol": "**",
      "provisioningState": "Succeeded",
      "resourceGroup": "rg-sentinel-test-events",

```

Delete the test network security group:

```
az network nsg delete \
  --resource-group rg-sentinel-test-events \
  --name nsg-sentinel-test
```

```
muhammad-ayman@fedora:~/Downloads/sentinel-bicep-homelab$ az network nsg delete \
> --resource-group rg-sentinel-test-events \
> --name nsg-sentinel-test
muhammad-ayman@fedora:~/Downloads/sentinel-bicep-homelab$
```

Delete the temporary group after the events have been generated:

```
az group delete \
  --name rg-sentinel-test-events \
```

```
--yes \  
--no-wait
```

```
muhammad-ayman@fedora:~/Downloads/sentinel-bicep-homelab$ az group delete \  
> --name rg-sentinel-test-events \  
> --yes \  
> --no-wait  
muhammad-ayman@fedora:~/Downloads/sentinel-bicep-homelab$
```

Do not create public-facing machines, malicious files, brute-force traffic, or exploit traffic for this starter lab.

12. Confirm the Network Detection

Wait several minutes, then run:

```
AzureActivity  
| where TimeGenerated > ago(30m)  
| where ResourceProviderValue =~ "Microsoft.Network"  
| where OperationNameValue has_any ("networkSecurityGroups", "securityGroups")  
| project TimeGenerated, Caller, OperationNameValue, ActivityStatusValue, ResourceGroup,  
ResourceId  
| order by TimeGenerated desc
```

The screenshot displays the Microsoft Defender Advanced Hunting interface. On the left is a navigation pane with categories like Home, Incidents, Exposure management, Investigation & response, Threat intelligence, Assets, Microsoft Sentinel, Identities, Endpoints, Email & collaboration, Cloud apps, Cloud security, SOC optimization, Reports, Learning hub, Trials, and More resources. The 'Advanced hunting' section is selected. The main area is titled 'Advanced hunting' and shows a query editor with the following query:

```
1 AzureActivity  
2 | where TimeGenerated > ago(30m)  
3 | where ResourceProviderValue =~ "Microsoft.Network"  
4 | where OperationNameValue has_any ("networkSecurityGroups", "securityGroups")  
5 | project TimeGenerated, Caller, OperationNameValue, ActivityStatusValue, ResourceGroup, ResourceId  
6 | order by TimeGenerated desc
```

Below the query editor, the 'Results' tab is active, showing 2 items. The results table has columns for TimeGenerated, Caller, OperationNameValue, ActivityStatusValue, ResourceGroup, and ResourceId. The results are as follows:

TimeGenerated	Caller	OperationNameValue	ActivityStatusValue	ResourceGroup	ResourceId
Aug 11, 2026 3:59...	ayman@globalbrandsgr...	MICROSOFTNETWORK/...	Accept	RG-SENTINEL-TEST-EVE...	
Aug 11, 2026 3:59...	ayman@globalbrandsgr...	MICROSOFTNETWORK/...	Start	RG-SENTINEL-TEST-EVE...	

Then open **Microsoft Sentinel > Incidents** and look for:

Lab - Network security group changed

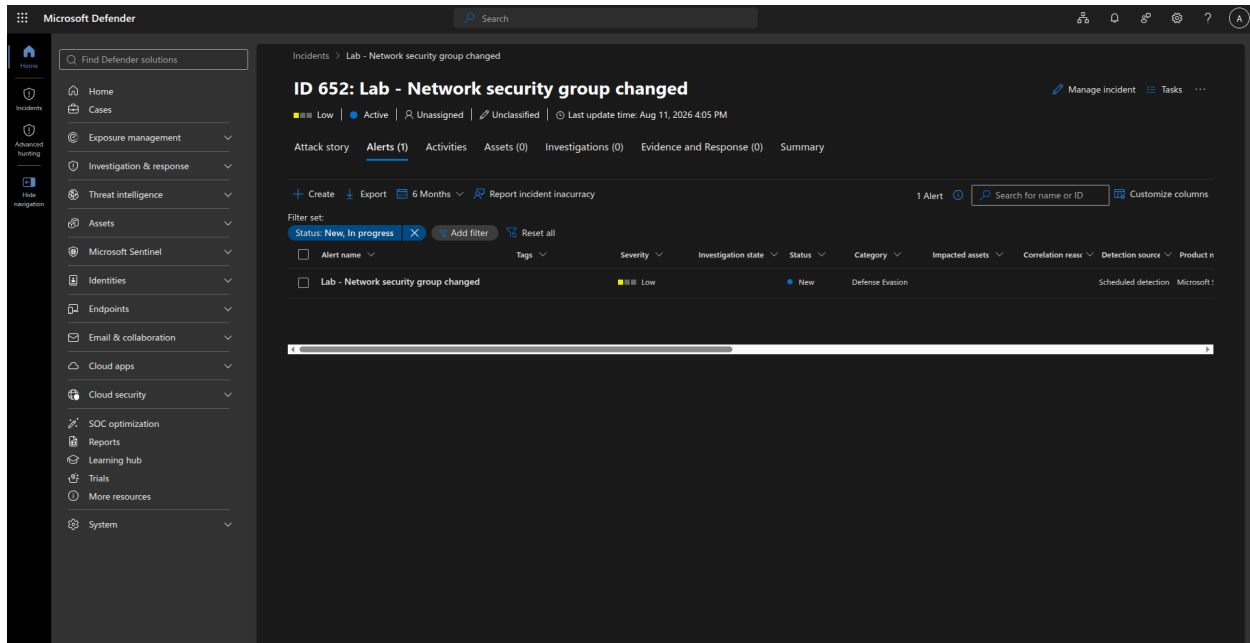
Open the incident and inspect the alert, entities, timeline, caller, operation name, and affected resource.

If the Activity Log query returns an event but no incident appears, check that the analytics rule is enabled and that its lookback window includes the event.

Scheduled rules can take another evaluation cycle to create an incident.

The screenshot shows the Microsoft Defender Incidents page. The left sidebar contains navigation options like Home, Incidents, Cases, Exposure management, Investigation & response, Threat intelligence, Assets, Microsoft Sentinel, Identities, Endpoints, Email & collaboration, Cloud apps, Cloud security, SOC optimization, Reports, Learning hub, Trials, More resources, and System. The main content area is titled 'Incidents' and includes a search bar, alert service settings, and email notification options. Below this, there are two summary cards: 'Attack disruptions' showing 'None' and 'Multi-alert incidents' showing '16%'. A table of incidents is displayed with columns for Incident name, Incident Id, Priority, Tags, Severity, Investigation state, Categories, and Impacted assets. The first incident, 'Lab - Network security group changed', is highlighted with a red box.

Incident name	Incident Id	Priority	Tags	Severity	Investigation state	Categories	Impacted assets
Lab - Network security group changed	652	1		Low		Defense evasion	
Lab - Azure Policy assignment changed	650	1		Medium		Defense evasion	
Lab - Azure Policy assignment changed	651	1		Medium		Defense evasion	



13. Clean Up the Lab

First remove the temporary test group if it still exists:

```
az group delete \
  --name rg-sentinel-test-events \
  --yes \
  --no-wait
```

```
muhammad-ayman@fedora:~/Downloads/sentinel-bicep-homelab$ az group delete \
> --name rg-sentinel-test-events \
> --yes \
> --no-wait
muhammad-ayman@fedora:~/Downloads/sentinel-bicep-homelab$
```

Delete the subscription-level Activity Log diagnostic setting. The Bicep deployment is not being used here to delete resources, so remove this explicitly:

```
az monitor diagnostic-settings subscription delete \
  --name send-activity-to-sentinel
```

```
muhammad-ayman@fedora:~/Downloads/sentinel-bicep-homelab$ az monitor diagnostic-settings subscription delete \
--name send-activity-to-sentinel
Are you sure you want to perform this operation? (y/n): y
muhammad-ayman@fedora:~/Downloads/sentinel-bicep-homelab$
```

Delete the lab resource group:

```
az group delete \
  --name rg-sentinel-homelab \
  --yes \
  --no-wait
```

```
muhammad-ayman@fedora:~/Downloads/sentinel-bicep-homelab$ az group delete \
--name rg-sentinel-homelab \
--yes \
--no-wait
muhammad-ayman@fedora:~/Downloads/sentinel-bicep-homelab$
```

14. Troubleshooting

Bicep validation fails

- Update Azure CLI and Bicep.
- Confirm the selected region supports Log Analytics and Microsoft Sentinel.
- Confirm your account has the required permissions.
- Check that resource names are unique where required.

Sentinel is not visible

- Confirm the workspace resource exists.
- Confirm the `onboardingStates/default` resource deployed successfully.
- Open Microsoft Sentinel from the workspace rather than searching only by resource group.
- Check the deployment operation details for authorization errors.

AzureActivity is empty

- Confirm the subscription diagnostic setting points to the deployed workspace.
- Confirm the enabled categories include `Administrative`.
- Wait 5 to 15 minutes after the first deployment.
- Generate a new test event after the diagnostic setting is deployed.

The query returns events but no incident

- Confirm the rule is enabled under **Analytics**.
- Check the rule's query frequency and lookback period.
- Verify the event matches the query exactly.
- Wait for at least one scheduled evaluation cycle.
- Check **Incidents** and **Alerts**, not only the Logs result.

Deployment creates an authorization error

- Ask the subscription administrator which role is missing.
- Remember that subscription diagnostic settings and Sentinel permissions may be separate from resource-group Contributor access.
- Do not grant permanent Owner access solely to complete a lab if a narrower role is available.

15. Next Exercises

After the starter lab works, extend it one feature at a time:

- Add a Microsoft Entra ID connector and test sign-in detections.
- Add a storage account and enable only the logs needed for a specific exercise.
- Add a small Linux VM with automatic shutdown, then monitor SSH authentication logs.
- Create a Logic App playbook with a managed identity for incident notification.
- Move analytics rules into separate Bicep modules.
- Add Azure Policy requiring `environment` and `managedBy` tags.

- Store the Bicep project in Git and deploy it from GitHub Actions or Azure DevOps.
- Add a CI pipeline that runs `az bicep build` and subscription deployment validation on every pull request.